

2026 Aug I
Module 1

Question 1.

[Tg:@abusat_760] Sphere A and Sphere B are tangent to each other at point P . Segment MP is a diameter of Sphere A, segment NP is a diameter of Sphere B, and point P lies on line segment MN . The radius of Sphere A is 5 times the radius of Sphere B. If the volume of Sphere B is 288π in.³, what is the length of segment MN in inches?

Question 2.

[Tg:@abusat_760] To cut a lawn, Antwan charges a fee of \$10.00 for his equipment and \$8.50 per hour spent cutting a lawn. Taylor charges a fee of \$7.00 for his equipment and \$9.50 per hour spent cutting a lawn. If x represents the number of hours spent cutting a lawn, what are the values of x for which Taylor's total charge is greater than Antwan's total charge?

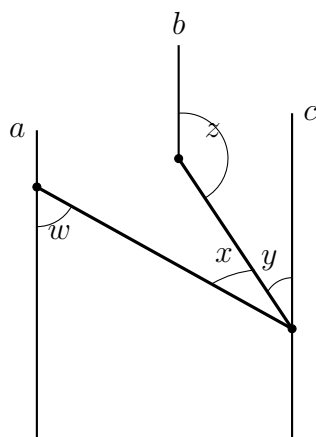
- A) $2 \leq x \leq 3$
- B) $3 \leq x \leq 4$
- C) $x < 2$
- D) $x > 3$

Question 3.

[Tg:@abusat_760] The relationship between the variables x and y is defined by an exponential equation. When $x = 0$ the value of y is 30, and for every increase in the value of x by 1, the corresponding value of y increases by 70% of its previous value. Which equation represents this relationship?

- A) $y = 30(1.70)^x$
- B) $y = 30(1.07)^x$
- C) $y = 70(1.30)^x$
- D) $y = 70(1.03)^x$

Question 4.



Note: Figure not drawn to scale.

[Tg:@abusat_760] In the figure shown, lines a , b , and c are parallel. If $201 < x + y + z < 212$, which of the following could be true?

- I. $w - y = 22$
- II. $w - y = 32$
- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

Question 5.

[Tg:@abusat_760] A circle has center O , and points P and Q lie on the circle such that segment PQ is a diameter of the circle. The length of arc PQ is 56 centimeters. What is the circumference, in centimeters, of the circle?

- A) 42
- B) 56
- C) 112
- D) 224

Question 6.

[Tg:@abusat_760] In 2004, Aster earned 11% more than in 2003, and in 2005 Aster earned 6% more than in 2004. If Aster earned y times as much in 2003 as in 2005, which of the following is closest to the value of y ?

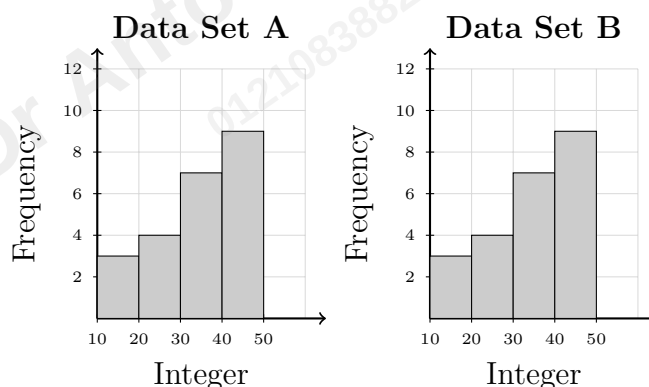
- A) 0.5455
- B) 0.6600
- C) 0.8409
- D) 1.1766

Question 7.

[Tg:@abusat_760] A community organization surveyed a random sample of residents to estimate the percentage of residents who visit their local library at least once per month. The organization estimates that 19% of residents visit the library at least once per month, with an associated margin of error of 3.7%. The organization repeated the survey with a random sample that was double the original sample size. Which of the following is true about the margin of error associated with the estimate from the larger sample?

- A) The margin of error is less than 3.7%.
- B) The margin of error is 7.4%.
- C) The margin of error is greater than 19%.
- D) The margin of error is 38%.

Question 8.



[Tg:@abusat_760] Two data sets of 23 integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to 10, but less than 20. The second interval represents the frequency of integers greater than or equal to 20, but less than 30, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

- A) 0
- B) 1
- C) 10
- D) 23

Question 9.

$$g(x) = (x + 14)(t - x)$$

[Tg:@abusat_760] The function g is defined by the given equation, where t is a constant. In the xy -plane, the graph of $y = g(x)$ passes through the point $(18, 0)$. What is the value of $g(0)$?

Question 10.

[Tg:@abusat_760] In the ancient Roman measurement system, a leuga was a unit of length that was equal to 7,500 pedes. In this measurement system, how many pedes were equivalent to 4.2 leugas?

Question 11.

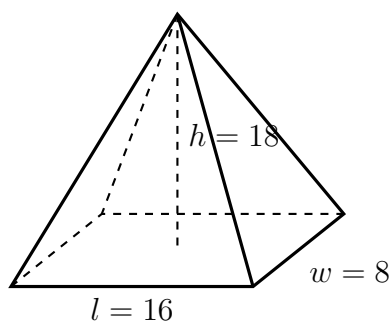
$$y = \frac{35 - x}{ax + b}$$

[Tg:@abusat_760] What is the x -intercept of the graph of the equation in the xy -plane, where a and b are positive constants?

- A) $(35, 0)$
- B) $(0, -35)$
- C) $(0, \frac{35}{b})$
- D) $(-\frac{b}{a}, 0)$

Question 12.

[Tg:@abusat_760] A triathlon is a multisport race consisting of three different consecutive events. A triathlon participant plans to complete the cycling event with an average speed of 31,680 yards per hour. What is this average speed, in miles per hour? (1 mile = 1,760 yards)

Question 13.

Note: Figure not drawn to scale.

[Tg:@abusat_760] The figure shown is a right rectangular pyramid, where $l = 16$ units, $w = 8$ units, and $h = 18$ units. What is the surface area, in square units, of the pyramid?

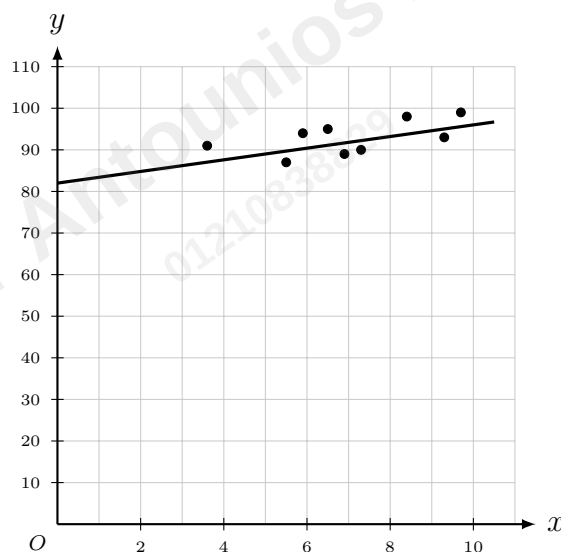
Question 14.

$$f(x) = 18(2.80)^{\frac{x}{2}}$$

[Tg:@abusat_760] The function f is defined by the given equation. The value of $f(x)$ increases by $p\%$ for every increase of x by 4. For which of the following functions, where n is a positive constant, does the value of $g(x)$ increase by $p\%$ for every increase of x by 1?

- A) $g(x) = n(1.40)^x$
- B) $g(x) = n(2.80)^x$
- C) $g(x) = n(6.84)^x$
- D) $g(x) = n(7.84)^x$

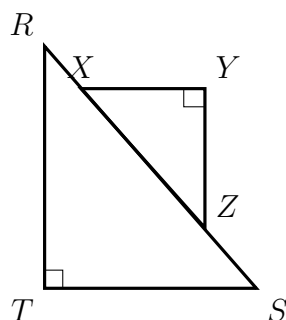
Question 15.



[Tg:@abusat_760] The scatterplot shows 9 data points and their line of best fit. For how many of the 9 data points is the actual y -value greater than the y -value predicted by the line of best fit?

- A) 8
- B) 7
- C) 5
- D) 2

Question 16.



[Tg:@abusat_760] In triangles RST and XYZ shown, $\overline{XY}$ is parallel to $\overline{TS}$ and $\tan R = \frac{160}{231}$. What is the value of $\sin X$ in triangle XYZ ?

- A) $\frac{160}{391}$
- B) $\frac{160}{281}$
- C) $\frac{231}{281}$
- D) $\frac{231}{160}$

Question 17.

$$f(x) = (x - 3)(x + 5)$$

[Tg:@abusat_760] The function f is defined by the given equation. The function g is defined by $g(x) = f(x - 2) + 4$. Which of the following equations is an equivalent form of $g(x)$ that displays the minimum value of g as a constant or coefficient?

- A) $g(x) = (x - 5)(x + 3) + 4$
- B) $g(x) = (x + 1)^2 - 12$
- C) $g(x) = (x - 1)^2 - 12$
- D) $g(x) = x^2 - 2x - 11$

Question 18.

[Tg:@abusat_760] A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species. At the end of the week, the number of prey had increased by 1900% of the number of prey at the start of the week, and the number of predators had increased by 220% of the number of predators at the start of the week. The number of predators at the end of the week was $p\%$ less than the number of prey at the end of the week. What is the value of p ?

Question 19.

[Tg:@abusat_760] Each 3.0-ounce serving of cheddar cheese and each 1.2-ounce serving of tuna provides about 1 microgram of vitamin B12. If a total of 3.5 micrograms of vitamin B12 are consumed from eating x ounces of cheese and y ounces of tuna, which equation best represents this situation?

- A) $0.33x + 0.33y = 3.5$
- B) $3.0x + 1.2y = 3.5$
- C) $1.2x + 3.0y = 3.5$
- D) $0.33x + 0.83y = 3.5$

Question 20.

[Tg:@abusat_760] In triangle RST , $RS = 504$, $ST = 128$, and $TR = 520$. Triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . What is the value of $\tan W$?

- A) $\frac{16}{65}$
- B) $\frac{16}{63}$
- C) $\frac{63}{65}$
- D) $\frac{63}{16}$

Question 21.

[Tg:@abusat_760] Which of the following data sets has the largest standard deviation, where p is a constant?

- A) $p - 10, p - 5, p, p - 2, p + 2$
- B) $p - 5, p - 5, p, p, p - 2$
- C) $p - 2, p + 2, p, p, p - 10$
- D) p, p, p, p, p

Question 22.

[Tg:@abusat_760] In the equation $7x(x + 11) = r$, r is a positive integer. How many distinct real solutions does the equation have for x ?

- A) Zero
- B) Exactly one
- C) Exactly two
- D) Infinitely many

Module 2

Dr Antounios Ashraf
01210838829

Question 1.

[Tg:@abusat_760] A rectangle is inscribed in a circle. If the length of the diagonal of the rectangle is $482\sqrt{3}$, what is the radius of the circle?

- A) $241\sqrt{3}$
- B) $482\sqrt{3}$
- C) $964\sqrt{3}$
- D) $241\sqrt{6}$

Question 2.

[Tg:@abusat_760] What is the sum of the solutions to the equation $x^2 - 90x - 18 = 0$?

- A) -90
- B) -18
- C) 18
- D) 90

Question 3.

[Tg:@abusat_760] In the xy -plane, the function $g(x)$ is defined as $g(x) = \frac{f(x)}{x+4}$. If the y -intercept of the graph of $y = g(x)$ is $(0, 54)$, what is the value of $f(0)$?

- A) 13.5
- B) 54
- C) 216
- D) 220

Question 4.

[Tg:@abusat_760] A square pyramid has a base side length of $a = 4$ units and a height of $h = 6$ units. What is the volume of the pyramid in cubic units?

- A) 8
- B) 16
- C) 32
- D) 48

Question 5.

[Tg:@abusat_760] The graph of a quadratic function $y = f(x)$ opens upward with its vertex located below the line $y = -9$. Which of the following could be the number of real solutions to the equation $f(x) = -9$?

- A) Zero
- B) Exactly one
- C) Exactly two
- D) Infinitely many

Question 6.

[Tg:@abusat_760] If $7(x - 4)^2 = 700$ and $x > 0$, what is the value of x ?

- A) 6
- B) 10
- C) 14
- D) 104

Question 7.

[Tg:@abusat_760] Line j passes through the points $(12, 0)$ and $(8, 0)$ in the xy -plane. Line k is perpendicular to line j and passes through the origin $(0, 0)$. At what x -coordinate do line j and line k intersect?

- A) -12
- B) 0
- C) 8
- D) 10

Question 8.

[Tg:@abusat_760] A quadratic function $f(x)$ satisfies $f(0) = 3$ and $f(8) = 3$. Which of the following equations could define $f(x)$?

- A) $f(x) = x(x - 8) + 3$
- B) $f(x) = x(x + 8) + 3$
- C) $f(x) = (x - 3)(x - 8)$
- D) $f(x) = 3x^2 - 8x$

Question 9.

[Tg:@abusat_760] In $\triangle ABC$, angle C is a right angle ($\angle C = 90^\circ$). Point E lies on side AC such that $CE = 2 \cdot AE$, and point D lies on side AB such that segment ED is perpendicular to AC . If $BC = 198$, what is the length of segment ED ?

- A) 66
- B) 99
- C) 132
- D) 396

Question 10.

$$f(x) = 16 - \frac{x}{27}$$

[Tg:@abusat_760] The function f is defined by the given equation. What is the value of $f(270)$?

- A) -10
- B) 6
- C) 10
- D) 26

Question 11.

$$\begin{cases} y = 6 \\ y = -4x + 2 \end{cases}$$

[Tg:@abusat_760] A system of linear equations is given. What is the solution (x, y) to the system?

- A) $(-1, 6)$
- B) $(1, 6)$
- C) $(6, -1)$
- D) $(6, 22)$

Question 12.

[Tg:@abusat_760] The total remaining distance, in miles, of a runner after t hours is modeled by the function $g(t) = 22 - 4t$. What is the best interpretation of the number 22 in this context?

- A) The speed of the runner in miles per hour
- B) The total time taken to finish the run
- C) The initial total distance, in miles, of the run
- D) The distance covered in the first hour

Question 13.

[Tg:@abusat_760] A quantity decreases by 20% every 5 seconds. If the initial value of the quantity is 340, which function $f(x)$ represents the value of the quantity after x seconds?

- A) $f(x) = 340(0.2)^{5x}$
- B) $f(x) = 340(0.8)^{5x}$
- C) $f(x) = 340(0.2)^{\frac{x}{5}}$
- D) $f(x) = 340(0.8)^{\frac{x}{5}}$

Question 14.

[Tg:@abusat_760] A company's annual revenue was x dollars in 2012. The revenue increased by 12% from 2012 to 2013, and increased by 4% from 2013 to 2014. If the revenue in 2014 was y times the revenue in 2012, which expression represents y ?

- A) $(1.12)(1.04)$
- B) $(1.12) + (1.04)$
- C) $\frac{1.04}{1.12}$
- D) $\frac{1}{(1.12)(1.04)}$

Question 15.

x	y
1	4,000
2	2,400
3	1,440
4	864

[Tg:@abusat_760] The table shows values of x and y . Which type of function best models the relationship between x and y ?

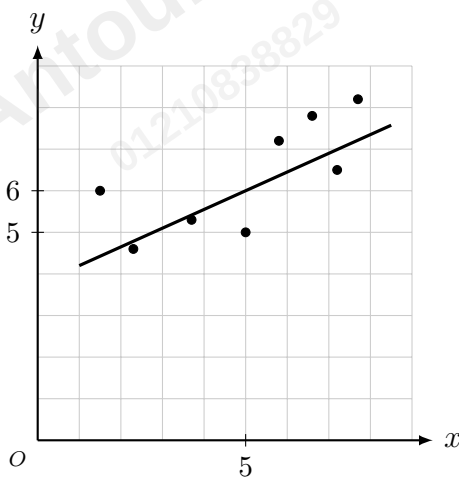
- A) Linear
- B) Exponential decay
- C) Quadratic
- D) Absolute value

Question 16.

[Tg:@abusat_760] During a triathlon, an athlete swims at a rate of 31,152 inches per hour. Given that 1 yard = 36 inches, what is this speed in yards per hour?

- A) 17.7
- B) 865.33
- C) 1,760
- D) 8,653.33

Question 17.



[Tg:@abusat_760] The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown. For $x = 5$, how much greater is the y -value predicted by the line of best fit than the actual y -value of the data point?

- A) 1
- B) 5
- C) 6
- D) 11

Question 18.

$$g(x) = -\sqrt{x^2 + bx + c}$$

[Tg:@abusat_760] The function g is defined by the given equation, where b and c are integers. In the xy -plane, the graph of $y = g(x)$ passes through the points $(2, 0)$ and $(0, -\sqrt{298})$. If m is a value for which $g(m) = 0$, what is the greatest possible value of m ?

- A) -150
- B) 2
- C) 149
- D) 150

Question 19.

$$18p - 19p = 17$$

[Tg:@abusat_760] What is the value of p that satisfies the equation?

- A) -17
- B) $-\frac{1}{17}$
- C) 17
- D) 37

Question 20.

[Tg:@abusat_760] Points A and B lie on a circle centered at point C . If the length of line segment AC is 34 mm, what is the length of segment BC ?

- A) 17 mm
- B) 34 mm
- C) 68 mm
- D) 115.6 mm

Question 21.

[Tg:@abusat_760] Which of the following expressions is equivalent to $\frac{4}{x-5} - \frac{4}{(x-5)^2}$ for all $x > 5$?

- A) $\frac{0}{(x-5)^2}$

B) $\frac{4x - 24}{(x - 5)^2}$

C) $\frac{4x - 16}{(x - 5)^2}$

D) $\frac{4}{x - 5}$

Question 22.

$$\begin{cases} 3x = 2 + 8y \\ -3x = -5 + 8y \end{cases}$$

[Tg:@abusat_760] Consider the system of linear equations above. What is the value of $6x$?

A) $\frac{7}{2}$

B) 7

C) 14

D) 21